Atharva Dwivedi

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LinkedIn | GitHub | HuggingFace | LeetCode

Professional Summary

AI Engineer proficient in building production-ready Large Language Model (LLM) applications, multi-agent systems, RAG pipelines, and deep learning solutions using Python, FastAPI, and modern AI frameworks. Strong foundation in machine learning, backend engineering, and distributed systems, with a passion for deploying reliable AI applications at scale.

Skills

- **Languages:** Python, Go, SQL, C++
- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Deep Learning, Computer Vision, Natural Language Processing (NLP), Reinforcement Learning
- **LLM & Generative AI:** Hugging Face Transformers, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, LlamaIndex, Prompt Engineering, Vector Embeddings, Fine-Tuning, LoRA/QLoRA
- **AI Deployment & MLOps:** FastAPI, REST APIs, vLLM, Docker, AWS (EC2, S3, RDS), GCP, Git, GitHub, CI/CD Pipelines, Weights & Biases (WandB)
- **Databases & Vector Stores:** PostgreSQL, SQLite, Redis (Caching & Pub/Sub), ChromaDB, FAISS, MongoDB
- **AI Developer Tools:** Claude Code, Cursor, GitHub Copilot, Antigravity
- **Core Engineering:** Data Structures & Algorithms, System Design, Distributed Systems, Microservices

Experience

Helionyex Solutions **June 2026 – Present**
Software Developer Intern *Remote*

- Developed core modules for a cross-platform **Climate Data Reception & Processing System (CDRPS)** using Electron, React, TypeScript, and PostgreSQL, designing secure authentication, RBAC rules, and station monitoring workflows.
- Designed and implemented a **real-time data quality validation engine** for meteorological observations, incorporating 10+ WMO-compliant QC rules to automatically catch anomalous readings before PostgreSQL storage.
- Architected high-speed local communications by developing **20+ secure IPC interfaces**, database transaction repositories, and internal authentication APIs to accelerate feature scaling.

Projects

ArchiGen AI – Multi-Agent Excalidraw Diagram Generator Website
Python, FastAPI, React, TypeScript, LangGraph, Excalidraw, Docker, Groq

- Engineered a **multi-agent AI pipeline** using **LangGraph** and **Groq** to autonomously layout software architecture diagrams, cutting manual drafting time by **90%** and rendering complex graphs in **under 8 seconds**.
- Implemented a **self-healing validation loop** with deterministic graph checks and LLM-based repair agents, eliminating rendering errors by **95%** while maintaining structural integrity across Excalidraw exports.
- Developed **real-time streaming infrastructure** using **FastAPI** Server-Sent Events (SSE), supporting **5 export formats** with a benchmarked **sub-100ms** streaming latency under concurrent user sessions.

AI-Enhanced Semantic File-Manager github.com/Atharv-3105/SmartFile-Manager
Go, Python, FastAPI, EasyOCR, Sentence-Transformers, SQLite, Docker

- Architected a real-time concurrent file indexing pipeline in Go supporting **8 concurrent ingestion workers**, indexing **500+ local documents** via an event-driven processing design.
- Built a Python extraction microservice using EasyOCR and Sentence-Transformers to generate **384-dimensional semantic embeddings**, enabling sub-**150 ms** cosine-similarity text retrieval.
- Engineered a containerized microservice architecture with Go, Python, SQLite, and Docker, implementing transactional storage, vector persistence, and automated local file system monitoring.

Semantic-Duel – Real-Time Multiplayer Word Battle Game github.com/Atharv-3105/Semantic-Duel
Go, WebSockets, REST APIs, NLP Embeddings, Docker

- Architected a **real-time multiplayer game backend** via **WebSockets** supporting **50+ concurrent players** with low ∼**150ms** state synchronization latency.
- Engineered an **NLP-based semantic scoring engine** using vector embeddings, processing **100+ similarity evaluations per match** for dynamic game state resolution.
- Designed a **modular microservices architecture** separating matchmaking, session state, and scoring logic with extensible REST APIs optimized for unified leaderboard integration.

Research Implementations

- Implemented state-of-the-art architectures from scratch using PyTorch: **Transformer, LLaMA 2, ViT, DDPM, U-Net, GANs, VAE, PPO, SAC, TD3, DDPG, MHA, MLA, GQA, LoRA, Qwen, MoEs, DPO**.

Education

Bachelor of Technology in Artificial Intelligence & Machine Learning **2022 – 2026**
Dr. A. P. J. Abdul Kalam Technical University, Lucknow, UP

Achievements & Certifications

- Solved **600+ Data Structures & Algorithms** problems on LeetCode.
- Contributed open-source datasets and deep learning models to the **Hugging Face Hub**.
- Mentored **150+ educators** on Python and Generative AI fundamentals as a Python & AI Trainer at NimbleQ.